FOR A ZETA-STABLE MODULE EITHER IT IS THE SIMPLE MODULE CONCENTRATED AT A VERTEX OR THE TWO MAPS OF THE COMPLEX ARE INJECTIVE AND SURJECTIVE

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Editorial note. The sentence between the counted statement and the counted proof (source line 1722) says that with the case W = 0 excluded the lemma gives the proof of the preceding proposition; it is the author's own linking sentence and is outside the counted spans.

Allowed standard prerequisites (editorial).

- The stable module (B,a,b) attached to a point of the moduli space, the simple module S_i concentrated at a vertex i of I^0, and the maps alpha_i and beta_i of the complex built from B, a, b. Set up in the same subsection; the author displays the complex explicitly before the counted statement (source lines 1700-1708). Fixes the two maps whose injectivity and surjectivity the statement is about, and the modules between which the homomorphisms in the proof are taken.
- The theta-functional of the subgraph I^0 and the fact that both the simple module S_i and the stable module have theta equal to zero. Used inside the counted proof at source lines 1727-1730. Makes every non-zero homomorphism between the two stable modules an isomorphism, which is the mechanism of the proof.

Lemma 0.1 (cf. [6, 4.7])**.** *Let $\zeta^\bullet$ as above. Fix $i \in I^0$. If (B, a, b) is $\zeta^\bullet$-stable, then either of the following hold*:

(a) *$W = 0$ (hence $a = b = 0$), and B is the simple module S_i.*
(b) *The map α_i is injective and β_i is surjective.*

Proof. This is a standard argument. The kernel of α_i gives a homomorphism from S_i to (B, a, b). Since both are $\zeta^\bullet$-stable and have the same $\theta_{\widehat{\zeta}^\bullet}$ (both are 0), the

Date: doi:10.3842/SIGMA.2009.003; CC BY 4.0.

homomorphism is either 0 or an isomorphism. Similarly the cokernel of β_i is the dual of the space of homomorphisms from (B, a, b) to S_i. Thus we have the same assertion. □

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